Feature Extraction via 3-D Block Characteristics Sharing for Hyperspectral Image Classification

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***Abstract*— Spectral–spatial information plays an essential role in hyperspectral image (HSI) classification compared to pure spectral information. However, the neighbor spectral–spatial information of a pixel tends to be mixed into other ground coverings due to various external factors such as the weather and sensor jitter, and mainstream HSI classification methods present low sensitivity for spatial information in this situation. This article proposes a novel feature extraction method via 3-D block characteristics sharing (3-D-BCS) for HSI classification that redefines spatial–spectral information of a local region based on a superpixel perspective to overcome the spectral–spatial weak assumptions in feature extraction that consists of the following steps. First, 3-D blocks are obtained by performing an oversegmentation method on the raw HSI. Then, instead of global operation, a 3-D block-based Gabor filter is applied to the principal components of an HSI to extract the textural features. Next, an average operation is conducted on each shape adaptive region to address the spatial weak assumption and Gaussian weight is introduced into each superpixel block to overcome the spectral weak assumption. Thus, 3-D characteristics sharing blocks can be constructed by reshaping the above three kinds of spectral–spatial feature. Finally, the majority-based support vector machine (SVM) classifier is utilized to determine the final class labels of HSI at the decision fusion level. Experiments performed on several real hyperspectral data sets with limited training samples show that the proposed 3-D-BCS method outperforms the other types of the classification method.**

***Index Terms*— Feature extraction, hyperspectral image (HSI) classification, spectral–spatial weak assumptions, superpixel segmentation.**

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I. Introduction

HYPERSPECTRAL images (HSI) captured by remote sensors have several hundreds of contiguous and narrow spectral bands. Their rich discriminative spectral characteristics and spatial information about the surface materials are preserved in these bands; therefore, HSI can be utilized to distinguish materials by identifying the spectral differences between different materials in various scenes. Due to this superiority, HSI plays an important role in a wide variety of remote sensing applications such as spectral unmixing [1]–[3], target detection [4]–[6], and classification [7]–[9].

Over the past decade, many researchers have designed different classifiers to solve the recognition problem of ground coverings in hyperspectral data. Among these, the focus has always been on pixel-level classifiers such as support vector machine (SVM) [10]–[12], sparse representation (SR) [13], [14], extreme learning machine (ELM) [15], [16] and neural networks [17]–[19]. Those methods have been proved to obtain a good performance in solving HSI classification problems. However, pixel-level classifiers of the supervised paradigm often face two major problems: 1) The pixel-level classifiers that present low sensitivity to high-dimensional images will generate noisy classification results and 2) the pixel-level classifiers that encountered a performance bottleneck due to the spectral information of a kind of material are often easily mixed in the real scene by that from different classes. To overcome the above problems, plentiful dimension reduction (e.g., feature extraction) and spectral–spatial classification methods (i.e., traditional method [20] and deep learning [21]) have been developed to further improve the interpretation accuracy and efficiency for ground coverings.

Feature extraction methods aim to map hyperspectral data from high- to low-dimensional feature space to enhance the separability among pixels and decrease the computational burden of data processing. High-dimensional HSIs often exist in the "Hughes" phenomenon in which classification accuracy deteriorates as the data dimension increases. Therefore, some important dimensionality reduction techniques are proposed to solve the problem in the domain of hyperspectral applications such as classification. For instance, Zhang *et al.* [22] introduced a stochastic search strategy inspired by the clonal

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selection theory in an artificial immune system for the dimensionality reduction of HSI. Furthermore, some classic methods of dimensionality reduction are widely used in hyperspectral data processing to remove redundant information and improve the classification performance such as principal component analysis (PCA) [23], independent component analysis (ICA) [24], probabilistic PCA (PPCA) [25], multidimensional scaling (MDS) [26], and average image fusion (AIF) [27]. The PCA algorithm can ensure that the main feature information of an HSI is retained in the form of a small quantity of important principal components (PCs) but it cannot emphasize the spectral features of interest after reducing the dimension. The ICA method makes the transformed components as independent as possible. However, the complexity of the ICA-based method increases the computational burden. By contrast, the MDS also serves as a classic dimension reduction method that emphasizes preserving local neighborhoods in the lower-dimensional space. In other words, the distance among pixel vectors after dimensionality reduction is similar to that in the original space.

In addition, spectral–spatial classification methods based on contextual information (i.e., fixed and self-adaptive) integration are favored by researchers due to their remarkable classification accuracy and efficiency. For a fixed window context information, a joint sparse representation (JSR) method was introduced into HSI classification in [28] to incorporate the contextual information of a test pixel into the sparse recovery optimization problem. Peng *et al.* [29] developed a maximum likelihood estimation (MLE)-based JSR (MLEJSR) model to replace the traditional quadratic loss function with an MLE-like estimator for measuring the joint approximation error. Both the theoretical and experimental results proved the effectiveness of the MLEJSR method, particularly in the case of large noise (i.e., outliers in the spatial neighborhood). In [30], a fixed window-based optimizing method of pixelwise classification results was proposed by exploiting edge-preserving filtering (EPF) to get smoothed probabilities. However, although spatial information based on a fixed window can improve the classification effect of ground coverings, it is difficult to significantly promote the classification accuracy since the avoidance of nonhomogeneous pixel confusions.

For self-adaptive context information, superpixel segmentation is introduced into HSI classification as a frequently used technology of spatial information extraction because the size and shape of superpixels can be changed adaptively according to the different spatial structures, which can help avoid the loss of spectral–spatial information and effectively improve the classification performance. For example, considering the correlation among pixels within each superpixel, Tu *et al.* [31] proposed superpixel correlation coefficient representation (SCCR) to exploit the spatial structure information of HSI. Jiang *et al.* [32] proposed a superpixel wise PCA (SuperPCA) approach to learn the intrinsic low-dimensional features of HSIs. Experiments on public data sets have proven that the SuperPCA model significantly outperforms the conventional PCA-based dimensionality reduction baselines for HSI classification. Moreover, in [33], a classification framework combining superpixel-level SR and multitask learning was proposed for HSI, which used the prior class-level sparsity for multiple-feature fusion, and the correlation and distinctiveness of pixels in a spatial local region. Jia *et al.* [34] developed a cascade superpixel regularized Gabor feature fusion (CSRGFF) method for HSI classification in which the two kinds of Gabor features, magnitude and phase, are jointly conducted and make full use of the discriminative information contained in Gabor features.

Although the above-mentioned classification methods based on self-adaptive context information have achieved pleasing classification accuracy compared to fixed window-based classifiers, the spectral–spatial weak assumptions (close pixels in spectral space have the same class, and all the pixels in a superpixel belong to the same label) of local context information is often ignored in the process of extracting spatial information. Recently, the phenomenon of local-region spectral—spatial weak assumptions was verified by [35] for HSI noisy label detection. Meanwhile, the superpixel-to-pixel weighting distance (SPWD) method in [35] has demonstrated effectiveness in noisy label detection in terms of improving the classification accuracy of classifiers by considering local-region spectral–spatial weak assumptions.

This article introduces a novel feature extraction method via 3-D block characteristics sharing (3-D-BCS) into HSI classification. The motivation behind this work is to integrate the shape-adaptive spatial structure information of the hyperspatial scene into the feature extraction process by designing superpixel guidance-based texture and spatial information extraction technology. Specifically, the proposed 3-D-BCS method considers and solves the weak assumptions of context information in terms of spatial and spectral aspects. The proposed 3-D-BCS method consists of the following steps. First, the low-dimensional images are obtained by performing the MDS algorithm on the original HSI, and then the over-segmentation algorithm is employed to divide low-dimensional HSI into nonoverlapping 3-D blocks. Second, instead of global operation, a local block (superpixel)-based Gabor filter is applied to the PCs of HSI to extract textural features. Third, an average operation is conducted to each shape adaptive region to address the spatial weak assumption, and the Gaussian weight is introduced to each superpixel block to overcome the spectral weak assumption. Thus, 3-D characteristics sharing blocks can be constructed by performing the three above-described filtering operations for each superpixel block. Finally, the majority-based SVM classifier is utilized to determine the final class labels of HSI at the decision fusion level. The key contributions of this article are summarized as follows:

1) A 3-D-BCS that considers spectral–spatial weak assumptions is introduced into HSI feature extraction for the first time. It is found that the classification performance (i.e., accuracy) can be effectively improved by utilizing the local density and Gaussian weight to, respectively, redefine spectral and spatial information within 3-D blocks.
2) We found that, instead of global operation, using a superpixel-based Gabor filter can achieve higher classification accuracy in our proposed feature extraction framework. In addition, three classic filtering methods (recursive filter, Gauss filter, and Gabor filter) are,

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respectively, discussed in feature extraction in terms of spatial texture consistency to provide a reference for subsequent studies.

The remainder of this article is organized as follows. Section II describes the related work, including briefly introducing the MDS dimensionality reduction method, entropy rate superpixel (ERS) segmentation algorithm, and Gabor filter. Section III introduces the proposed 3-D-BCS method and majority-based SVM in detail. Section IV presents experimental results and discussions on real HSI data sets. Section V gives conclusions and future work.

II. Related Work

A. Multidimensional Scaling

MDS [26] as a minimum distortion unsupervised embedding method has been widely applied in many fields of research such as psychology, computer vision, and remote sensing. For the process of reducing dimension in hyperspectral data, we assume δ_2 is the distortion criterion and try to find a minimum distortion embedding into $\mathbb{R}^m$ by the generation function f as follows:

$$f = \underset{f:\mathbf{A}\to\mathbb{R}^m}{\arg\min} \sum_{i>j} \left| \mathrm{Dist}_{\mathbb{R}^m}\left(f(\mathbf{a}_i), f\left(\mathbf{a}_j\right)\right) - \mathrm{Dist}_{\mathbf{A}}\left(\mathbf{a}_i, \mathbf{a}_j\right) \right|^2 \quad (1)$$

where $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n]$ refers to pixels in the HSI. Let us assume $z_i = f(\mathbf{a}_i)$ represents the m-dimensional Euclidean coordinates of pixel $\mathbf{a}_i$ in the map under f, and the coordinates matrix $\mathbf{Z} = [z_1, z_2, \ldots, z_n] \in \mathbb{R}^{n\times m}$ can be easily obtained. The distortion criterion can be represented as follows:

$$\delta_2(\mathbf{Z}, \mathrm{Dist}(\mathbf{A})) = \sum_{i>j} |\mathrm{Dist}_{ij}(\mathbf{Z}) - \mathrm{Dist}_{ij}(\mathbf{A})|^2 \quad (2)$$

where $\mathrm{Dist}_{ij}(\mathbf{A})$ is the geodesic distances between the ith and jth pixel and $\mathrm{Dist}_{ij}(\mathbf{Z})$ denotes the Euclidean distance between the ith and jth pixel on the canonical form $\mathrm{Dist}_{\mathbb{R}^m}(z_i, z_j) = ||z_i - z_j||_2$. The rows of the embedding matrix Z are given by the leading eigenvectors of a linear transformation matrix that achieves the minimum distortion of (3). The embedding coordinates of the pixels can also be found as the solution to a nonlinear least-squares problem

$$\mathbf{Z}^* = \underset{\mathbf{Z}\in\mathbb{R}^m}{\arg\min}\, \delta_2(\mathbf{Z}). \quad (3)$$

B. Superpixel Segmentation

Superpixel segmentation algorithms have been widely used in computer vision applications such as image segmentation [36], target tracking [37], and target recognition [38]. A large number of pixels in an image are replaced by a small number of superpixels, which greatly reduces the complexity of image post-processing. In this article, superpixels are generated using the ERS [39] segmentation algorithm to obtain the picture's salient features. The superpixel generation function can be described as follows:

$$\mathbf{I} = \bigcup_{i=0}^{S_N} \mathbf{Y}_i \quad (4)$$

where S_N is the number of superpixels, and $\mathbf{Y}_i$ represents to the ith superpixel. $\mathbf{Y}_i \cap \mathbf{Y}_j = \varnothing$, there are no identical pixels inside two randomly selected superpixels.

The details of the ERS segmentation algorithm are as follows. To solve the problems caused by the different sizes and complex shapes of ground objects, the image is mapped onto a graph G as follows:

$$G = (V, E) \quad (5)$$

where V is the vertex set corresponding to the pixels of the image, and E is the edge set. An entropy rate term $H(\cdot)$ and a balancing term $B(\cdot)$ are introduced into the ERS segmentation algorithm, which can be formulated as follows:

$$H(O) = -\sum_i \mu_i \sum_j p_{i,j}(O) \log\left(p_{i,j}(O)\right) \quad (6)$$

$$B(O) = H(Z_O) - N_O = -\sum_i pz_O(i) \log(pz_O(i)) - N_O \quad (7)$$

where μ_i is a stationary distribution, and $p_{i,j}$ is the transition probabilities. N_O represents the number of connected components in the graph. Z_O refers to the distribution of the cluster membership. Then, the objective function of superpixel segmentation can be represented as follows:

$$\max_O \{H(O) + \mu B(O)\} \quad \text{s.t } O \subseteq E \quad (8)$$

where μ is the weight controlling the contributions of the entropy rate term and balancing term.

C. Gabor Filter

The Gabor filter [40], [41] is modulated by a Gaussian function and a sinusoidal plane wave. The texture and orientation information of an image can be extracted from Gabor-filtered images. The generating function of the Gabor filter can be represented as

$$\Phi_{u,v}(x, y) = \frac{c^2}{\pi\gamma\eta} \exp\left(-\left(\alpha^2 x'^2 + \beta^2 y'^2\right)\right) \exp\left(j2\pi c x'\right) \quad (9)$$

$$x' = \left(x - \frac{m+1}{2}\right)\cos\theta + \left(y - \frac{n+1}{2}\right)\sin\theta \quad (10)$$

$$y' = -\left(x - \frac{m+1}{2}\right)\sin\theta + \left(y - \frac{n+1}{2}\right)\cos\theta \quad (11)$$

where c represents the distribution coefficient of the frequency domain in the generating function and θ refers to the rotation angle of the Gaussian function. $\gamma = c/\alpha$ and $\eta = c/\beta$ are defined as the constraint conditions keeping the ratio between the frequency and sharpness and α is the sharpness of the Gaussian function along the major axis parallel to the wave and β is the sharpness of the Gaussian function along the short axis perpendicular to the wave. Here, α and β are assumed to be equal and $\gamma = \eta = \sqrt{2}$ was adopted in most previous studies [42]. In addition, m and n represent the size of the Gabor filter, which is usually set equal to an odd number.

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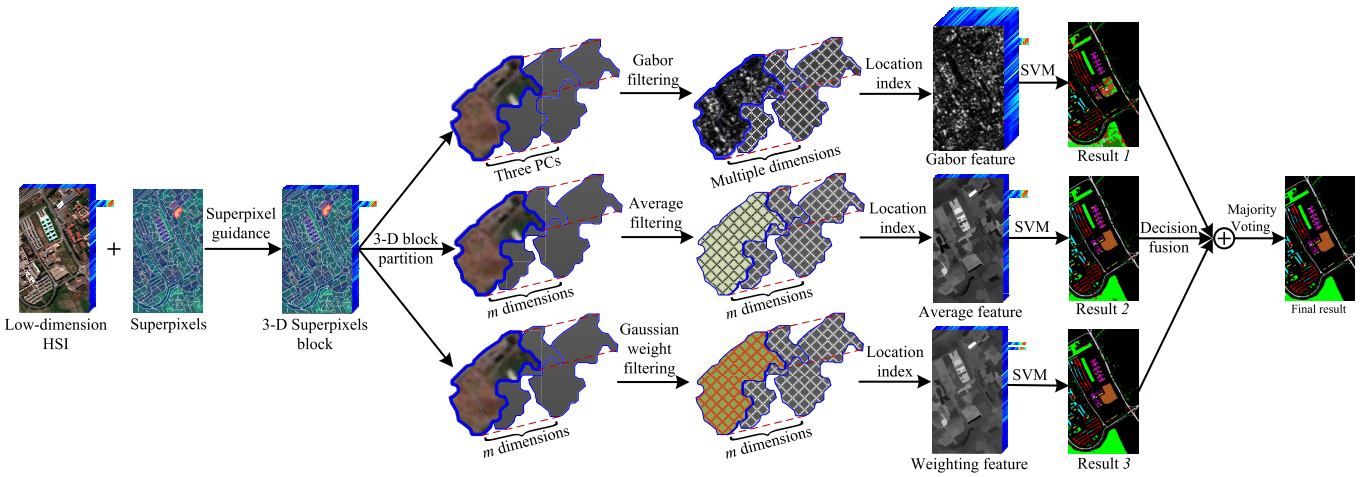


Fig. 1. Overview of the proposed 3-D-BCS method for feature extraction prior to HSI classification.

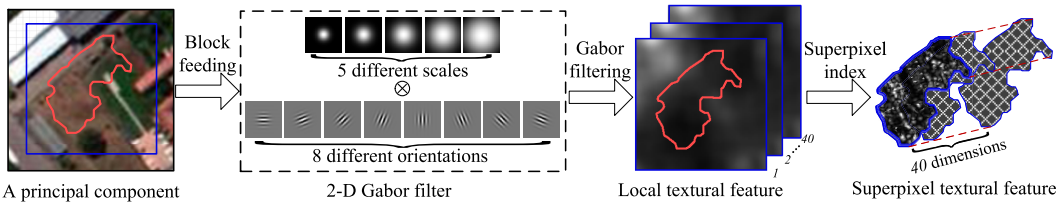


Fig. 2. Schematic of the superpixel-based Gabor filtering for the proposed method.

III. Proposed Approach

As shown in Fig. 1, the proposed feature extraction and classification approach[1] consists of three steps: 1) MDS-based 3-D blocks partitioning; 2) 3-D-BCS-based feature extraction; and 3) majority voting-based decision fusion. It is worth mentioning that, 3-D-BCS-based feature extraction can effectively overcome spectral–spatial weak assumptions by employing local density and Gaussian weight to, respectively, redefine spectral and spatial information within 3-D blocks compared to other feature extraction methods. Algorithm 1 illustrates the pseudocode of the proposed 3-D-BCS method. Each component contains some key steps, the details of which are as follows:

A. MDS-Based 3-D Blocks Partitioning

Although MDS and PCA are similar in data dimensionality reduction, MDS can make the distance between samples in low-dimensional space and the similarity between samples in high-dimensional space as consistent as possible. Let us assume that $\boldsymbol{I} = \{\boldsymbol{x_1}, \boldsymbol{x_2}, \ldots, \boldsymbol{x_N}\} \in \mathbb{R}^{N\times M}$ represents an original HSI, N is the number of pixels, and M is the dimensions of the original image. First, the MDS algorithm is employed to reduce the dimensions of the original HSI to decrease the computational burden and weaken the Hughes phenomenon. According to the theory of Section II-B, the formula is defined as follows:

$$\boldsymbol{D} = \mathrm{MDS}(\mathbf{I}, \mathrm{Q}) \tag{12}$$

[1]Available online: https://github.com/chengle-zhou

where $\boldsymbol{D} = \{\boldsymbol{r_1}, \boldsymbol{r_2}, \ldots, \boldsymbol{r_N}\} \in \mathbb{R}^{N\times Q}$ represents the dimension-reduced image, and Q is the dimensions of $\boldsymbol{D}$. Then, the former three dimensions $\boldsymbol{D_{3d}}$ of the $\boldsymbol{D}$ are used as three channels of red, green, and blue to synthesize a pseudo-color image $\boldsymbol{D_p}$. Meanwhile, $\boldsymbol{D_p}$ can be an input of the ERS algorithm to generate a nonoverlapping 2-D superpixel map $\boldsymbol{I_s}$, the formula of which is expressed as follows:

$$\boldsymbol{I}_s = \mathrm{ERS}_{S_N}(\boldsymbol{D_p}). \tag{13}$$

Finally, the 3-D superpixel block can be divided by mapping the edge detail of $\boldsymbol{I_s}$ into the reduced image $\boldsymbol{D}$.

B. Block Characteristics Sharing-Based Feature Extraction

This section briefly describes 3-D-BCS-based feature extraction. Specifically, spectral–spatial weak assumptions within a local region (i.e., a 3-D superpixel block) are considered and effectively overcome. The details of block characteristics sharing-based feature extraction are as follows:

1) *Superpixel-Based Gabor Filter:* As shown in Fig. 2, in spatial domain, the minimum envelope region $\boldsymbol{R_Q}$ of a superpixel is first extracted based on its boundary width. Then, the Gabor features $\mathbf{G}_{u,v}$ of the minimum envelope region $\boldsymbol{R_Q}$ can be obtained by, respectively, convolving the first three PCs $\boldsymbol{D_{3d}}$ with a family of Gabor filters $\Phi_{u,v}(c_x, c_y)$ as follows:

$$\boldsymbol{G}_{u,v}(c_x, c_y) = |\boldsymbol{R_Q} * \Phi_{u,v}(c_x, c_y)| \tag{14}$$

where $G_{u,v}(c_x, c_y)$ represents the Gabor features that are the convolution result that corresponds to the Gabor filter

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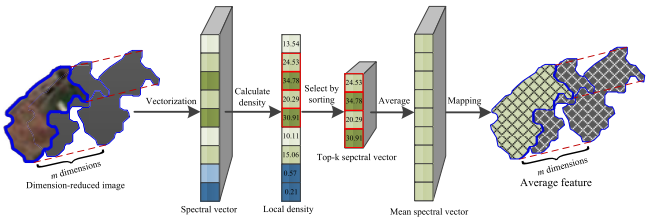


Fig. 3. Outline of the superpixel-based K-means filtering for the proposed method.

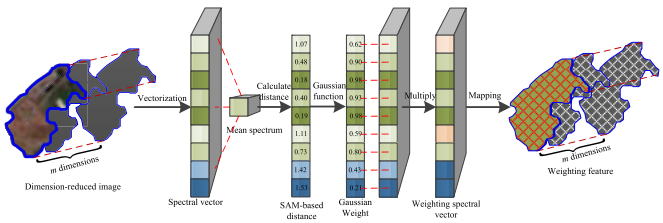


Fig. 4. Illustration of the superpixel-based Gaussian weighting filtering for the proposed method.

at scale v and orientation u. In this work, v and u are set to five and eight, respectively. Here, the reason for using local block-based Gabor filtering (GrF) is that local operations make the details inside a superpixel smoother than global operations, thus ensuring the texture consistency in the superpixel.

2) *Superpixel-Based K-Means Filter:* As shown in Fig. 3, the local density ρ_i of the ith sample in a superpixel can be first calculated according to density peak clustering (DPC) algorithm [43] as follows:

$$\rho_i = \sum_j \chi\left(d_{ij} - d_c\right) \tag{15}$$

where d_{ij} is the spectral angle [7] between the ith and jth samples in the cth superpixel, and the cutoff distance d_c represents the radius of the search region. If $d_{ij} > d_c$, $\chi(d_{ij} - d_c) = 0$. Otherwise, $\chi(d_{ij} - d_c) = 1$.
Thus, the local density set of all samples in the cth superpixel can be constructed as $\boldsymbol{\rho^c} = [\rho_1^c, \rho_2^c, \ldots, \rho_n^c]$. Then, to address the spatial weak assumption within the superpixel, the top K samples with the greatest confidence coefficient are selected based on the descending order of the pixels' local density. Finally, the average spectrum can be obtained by conducting the mean operation on the top K maximum spectral set $\tilde{\boldsymbol{r}}^{\boldsymbol{c}} = \{\tilde{\boldsymbol{r}}_{\boldsymbol{1}}^{\boldsymbol{c}}, \tilde{\boldsymbol{r}}_{\boldsymbol{2}}^{\boldsymbol{c}}, \ldots, \tilde{\boldsymbol{r}}_{\boldsymbol{K}}^{\boldsymbol{c}}\}(K < n)$ as follows:

$$\boldsymbol{J_c} = \frac{1}{K}\sum_{i=1}^{K} \tilde{\boldsymbol{r}}_{\boldsymbol{i}}^{\boldsymbol{c}} \tag{16}$$

where $K = k_0 \cdot n$ is the number of samples in set $\tilde{\boldsymbol{r}}^{\boldsymbol{c}}$, k_0 represents the ratio of selected samples in the superpixels, and n refers to the number of pixels within the suerpixel block. Then, the features of the samples in the cth superpixel are replaced by the average spectrum $\boldsymbol{J_c}$. Finally, all of the K-means features $\{\boldsymbol{J_1}, \boldsymbol{J_2}, \ldots, \boldsymbol{J_{S_N}}\}$ can be obtained by conducting the same operation on each superpixel block.

3) *Superpixel-Based Gaussian Weighting Filter:* As shown in Fig. 4, to overcome the spectral weak assumption among pixels within a superpixel, we introduced the Gaussian weight to redefine the proportion of the spectral vector within original superpixel based on the above average spectral vector. First, the spectral angle mapper (SAM)-based distance information between the average spectrum $\boldsymbol{J_c}$ and other pixels of the cth superpixel is obtained by

$$\sigma_i^c = \cos^{-1}\left(\frac{\boldsymbol{J}_c^T \cdot \boldsymbol{r}_i^c}{||\boldsymbol{J}_c|| \cdot ||\boldsymbol{r}_i^c||}\right). \tag{17}$$

Then, the Gaussian function is employed to compute the weight of the spectral vector, which is defined as follows:

$$\omega\left(\boldsymbol{r}_i^c\right) = \phi \cdot \exp\left(-\left(\sigma_i^c - \mu\right)^2 / 2\delta^2\right) \tag{18}$$

where ϕ refers to the height of the Gaussian function, μ is the centerline offset of the Gaussian function, and δ represents the half-peak width that controls the attenuation of weighting. Note that, ϕ and μ are set to one and zero, respectively. Next, a transformation of the spectral vector can be achieved by multiplying the original spectral vector by the weight coefficient. The formula is as follows:

$$\hat{\boldsymbol{r}}_{\boldsymbol{i}}^{\boldsymbol{c}} = \omega\left(\hat{\boldsymbol{r}}_{\boldsymbol{i}}^{\boldsymbol{c}}\right) \cdot \hat{\boldsymbol{r}}_{\boldsymbol{i}}^{\boldsymbol{c}}. \tag{19}$$

Finally, the weighted spectral vector of the cth superpixel can be represented as $\hat{\boldsymbol{r}}^{\boldsymbol{c}} = \{\hat{\boldsymbol{r}}_{\boldsymbol{1}}^{\boldsymbol{c}}, \hat{\boldsymbol{r}}_{\boldsymbol{2}}^{\boldsymbol{c}}, \ldots, \hat{\boldsymbol{r}}_{\boldsymbol{n}}^{\boldsymbol{c}}\}$, and the weighted spectral vector of other superpixels can also be calculated by conducting the same operation on each superpixel block.

C. Majority Voting-Based Decision Fusion

The SVM classifier is, respectively, utilized for the classification of the superpixel-based Gabor feature, superpixel-based K-means feature and superpixel-based Gaussian weighting features with a small training set. The class probability $P(c_j = l|\boldsymbol{r_i})$ of sample $\boldsymbol{r_i}$ that belongs to class l can be calculated from three kinds of classification result $\{c_1, c_2, c_3\}$ as follows:

$$P\left(c_j = l|\boldsymbol{r_i}\right) = \frac{\xi\left(c_j = l|\boldsymbol{r_i}\right)}{F_n} \tag{20}$$

where $\xi(c_j = l|\boldsymbol{r_i})$ represents the number of votes of class l, l is a subset of the class label set $\{1, 2, \ldots, L\}$, L is the total number of classes, and F_n is set to three in this work; therefore, the final class label of the sample $\boldsymbol{r_i}$ is determined by

$$\text{Class}(\boldsymbol{r_i}) = \arg\max_{l \in (1,L)} P(c_i = l|\boldsymbol{r_i}) \tag{21}$$

where $\text{Class}(\boldsymbol{r_i})$ refers to the majority voting-based classification result of $\boldsymbol{r_i}$.

IV. Experiments

A. Experimental Setup

1) Data Sets: For this section, experiments were carried out on the Indian Pines, Salinas Valley, University of Pavia, and

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TABLE I

NUMBER OF TRAINING AND TEST SAMPLES IN THE INDIAN PINES, SALINAS VALLEY, UNIVERSITY OF PAVIA, AND UNIVERSITY OF HUSTON DATA SETS, RESPECTIVELY

Indian Pines			Salinas Valley			University of Pavia			University of Houston		
Class	Training	Test	Class	Training	Test	Class	Training	Test	Class	Training	Test
Alfalfa	4	42	Weeds_1	5	2004	Asphalt	19	6612	Healthy_G	10	1063
Corn_N	15	1413	Weeds_2	9	3717	Meadows	19	18630	Stressed	10	800
Corn_M	9	821	Fallow	5	1971	Gravel	19	2080	Synthetic	10	687
Corn	4	233	Fallow_P	4	1390	Trees	19	3045	Trees	10	1043
Grass_M	7	476	Fallow_S	5	2673	Metal Sheets	19	1326	Soil	10	1232
Grass_T	7	723	Stubble	8	3951	Bare soil	19	5010	Water	10	315
Grass_P	2	26	Celery	7	3572	Bitumen	19	1311	Residential	10	968
Hay_W	5	473	Grapes	20	11251	Self-Blocking Bricks	19	3663	Commercial	10	614
Oats	3	17	Soil	11	6192	Shadow	19	928	Road	10	1021
Soybean_N	10	962	Corn	4	3274				Highway	10	372
Soybean_M	23	2432	Lettuce_4wk	4	1064				Railway	10	104
Soybean_C	6	587	Lettuce_5wk	4	1923				Parking_1	10	1223
Wheat	3	202	Lettuce_6wk	3	913				Parking_2	10	439
Woods	12	1253	Lettuce_7wk	3	1067				Tennis court	10	418
Buildings	5	381	Vinyard_U	13	7255				Runing track	10	650
Stone	4	89	Vinyard_T	3	1804						

Algorithm 1 3-D-BCS

Inputs: Original HSI $\boldsymbol{I} = \{\boldsymbol{x_1}, \boldsymbol{x_2}, \ldots, \boldsymbol{x}_N\} \in \mathbb{R}^{N\times M}$, Training set, and parameters setting: Q, S_N, K, and δ.

Outputs: Classification results obtained by decision fusion-based SVM classifier.

1: Obtain low-dimension HSI $\boldsymbol{D} = \{\boldsymbol{r_1}, \boldsymbol{r_2}, \ldots, \boldsymbol{r_N}\} \in \mathbb{R}^{N\times Q}$ by performing the MDS algorithm.
2: Generate nonoverlapping 3-D superpixel blocks by mapping $\boldsymbol{I_s}$ into the reduced image $\boldsymbol{D}$.
3: Block characteristic sharing-based feature extraction for each 3-D superpixel block.
4: **For** *each* $i = 1 : S_N$ **do**.
5: *Superpixel-based Gabor Filter*: Ensuring the texture consistency of each superpixel according to (14).
6: *Superpixel-based K-means Filter*: To address the spatial weak assumption within a superpixel, local density peak (15) and K nearest neighbor (16) are introduced.
7: *Superpixel-based Gaussian Weighting Filter*: Overcoming the spectral weak assumption among pixels within a superpixel according to (19).
8: **End For**
9: Superpixel Gabor feature-based SVM results.
10: Superpixel K-means feature-based SVM results.
11: Superpixel weighting feature-based SVM results.
12: Majority voting-based decision fusion for the classification results of the SVM classifier.

University of Houston data sets. The information from these data sets is presented as follows:

1) *Indian Pines Image:* The Indian Pines data set is from the Indian Pines test site in northwest Indiana, which was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. The image has a spatial dimension of 145 × 145 and 220 spectral bands with a spatial resolution of 20 m per pixel (20 water absorption channels were removed before experiments). As the scene was captured in June, certain crops such as corn and soybean were still in the early stage of growth. Fig. 5(a)–(c) shows the three-band color composite image, the corresponding reference image, and color-coding, respectively. In the reference classification map, the scene is divided into 16 classes.
2) *Salinas Valley Image:* The Salinas Valley data set was collected by the AVIRIS sensor over the Salinas Valley in California; it has 224 bands of size 512 × 217 pixels. In the experiments, 20 water absorption and noisy bands (no.108–112, 145–167, and 224) were removed. Fig. 6(a)–(c) illustrates the three-band color composite image, the corresponding reference image, and the corresponding color-coding of 16 ground coverings, respectively.
3) *University of Pavia Image:* The University of Pavia data set was obtained over the University of Pavia photographed by the Reflective Optics System Imaging Spectrometer (ROSIS-3). The image is of size 610 × 340 pixels with a spatial resolution of 1.3 m per pixel and 115 bands. After removing 12 noise bands, the experiments were conducted on the remaining 103 bands. Fig. 7(a)–(c) gives the false-color composite image of the University of Pavia data set, the corresponding reference date, and color-coding, respectively.
4) *University of Houston Image:* The University of Houston data set[2] was acquired over the Houston University campus and its neighboring area and was used in the 2013 Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest. The hyperspectral data contains 144 spectral bands in the 380–1050 nm region and 349 × 1905 pixels with a spatial resolution of 2.5 m. This image is an urban data set in which most of the land covers are man-made objects that contain 15 classes. Fig. 8 represents the false-color composite image of the

[2]http://www.grss-ieee.org/community/technical-committees/data-fusion/

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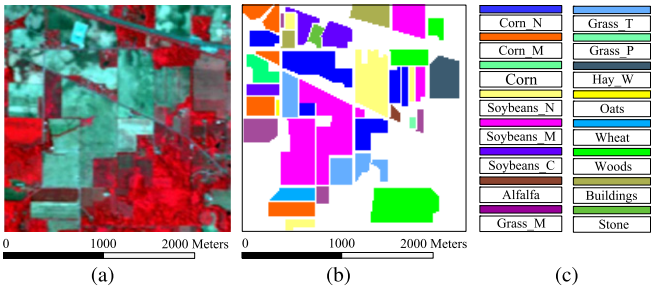


Fig. 5. Indian pines image. (a) Three-band color composite image. (b) Reference image. (c) Color-coding.

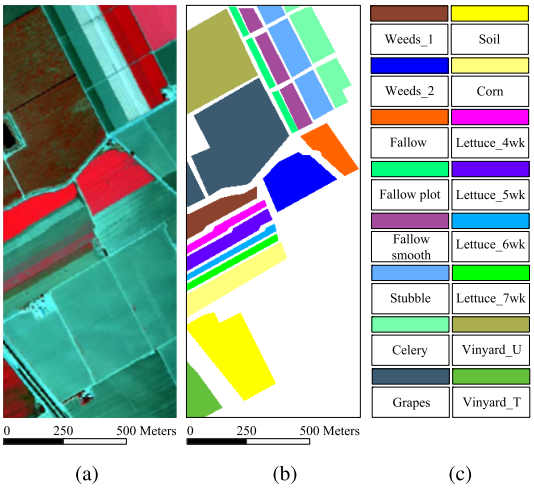


Fig. 6. Salinas valley image. (a) Three-band color composite image. (b) Reference image. (c) Color-coding.

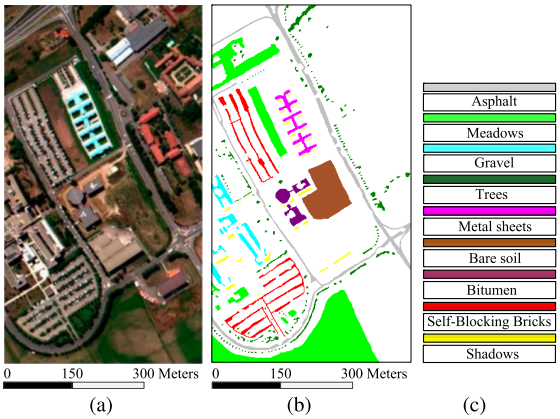


Fig. 7. University of Pavia image. (a) Three-band color composite image. (b) Reference image. (c) Color-coding.

Houston University image, the corresponding reference data, and color-coding, respectively.

2) Quality Indexes: In this article, three widely used quality indexes, the overall accuracy (OA), the average accuracy (AA), and the Kappa coefficient (Kappa) are calculated by averaging the results achieved after 10 repeated experiments with various randomly selected training sets. OA represents the percentage of pixels that are correctly classified, AA refers to the average percentage of correctly classified pixels for a different class. The Kappa coefficient, which measures the percentage of correctly classified pixels corrected by the number of agreement, is introduced to account for uncertainty factors in the classification. In addition, the average and standard deviation of these quality indexes are given in our experiments.

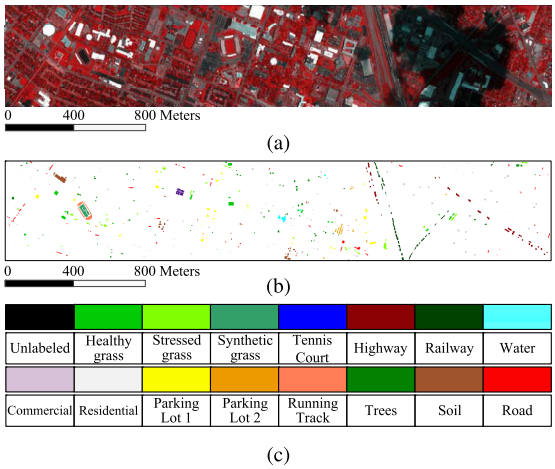


Fig. 8. University of Houston image. (a) Three-band color composite image. (b) Reference image. (c) Color-coding.

TABLE II

CLASSIFICATION ACCURACIES (IN %) OF THE 3-D-BCS METHOD USING DIFFERENT DIMENSIONALITY REDUCTION METHODS ON THE DIFFERENT REAL HSI DATA SETS. THE NUMBERS IN PARENTHESES REPRESENT THE STANDARD VARIANCE OF THE ACCURACIES OBTAINED ACROSS TEN REPEATED EXPERIMENTS

Indian Pines				
Indicator	AIF [27]	PCA [23]	PPCA [25]	MDS [26]
OA	75.92(2.06)	86.26(2.42)	77.54(2.74)	**87.78(2.31)**
AA	80.36(2.68)	88.37(2.21)	72.06(3.39)	**89.98(1.38)**
Kappa	72.52(2.37)	84.22(2.76)	74.40(3.06)	**85.92(2.71)**
Time(s)	11.82	10.82	604.6	10.73
Salinas Valley				
Indicator	AIF [27]	PCA [23]	PPCA [25]	MDS [26]
OA	92.78(2.79)	92.25(1.88)	94.68(1.67)	**95.73(1.76)**
AA	95.32(1.91)	95.86(1.43)	95.45(1.94)	**97.06(1.09)**
Kappa	91.91(3.15)	91.32(2.11)	94.07(1.85)	**95.23(1.97)**
Time(s)	14.82	15.07	1356.02	14.78
University of Pavia				
Indicator	AIF [27]	PCA [23]	PPCA [25]	MDS [26]
OA	85.45(1.67)	90.13(1.57)	85.20(3.39)	**91.66(2.73)**
AA	85.42(1.80)	88.59(1.91)	84.39(1.77)	**91.08(2.64)**
Kappa	81.17(2.12)	87.65(2.03)	80.75(4.25)	**89.00(3.47)**
Time(s)	32.99	32.79	1224.80	32.13
Houston of Pavia				
Indicator	AIF [27]	PCA [23]	PPCA [25]	MDS [26]
OA	73.76(0.74)	74.35(4.20)	73.75(3.71)	**76.73(1.81)**
AA	78.16(1.76)	80.32(1.45)	80.55(1.05)	**81.73(0.95)**
Kappa	71.63(0.79)	72.27(4.57)	71.63(4.00)	**74.83(1.95)**
Time(s)	177.19	174.73	5082.42	130.37

B. Analysis of the Influence of Different Dimensionality Reduction Tools

In this section, an experiment is performed on the above four HSI data sets to analyze the influence of various dimensionality reduction tools, such as AIF [27], PCA [23], PPCA [25], and MDS [26] methods, on the proposed 3-D-BCS method. In this experiment, to be objectively fair, the training samples of each data set are selected randomly (see Table I) in each iteration of the experiment. As shown in Table II, for all of the data sets, the 3-D-BCS method that adopts the MDS dimension reduction tool achieves the best classification performance

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TABLE III

CLASSIFICATION ACCURACIES (IN %) OF THE 3-D-BCS METHOD USING DIFFERENT FILTERING METHODS ON THE VARIOUS HSI DATA SET. NOTE THAT, S_N IS THE NUMBER OF SUPERPIXEL BLOCK TO BE FED, B_n IS MARKED AS THE NUMBER OF SUPERPIXEL BLOCK IN THE LARGE SCALE. $4B_n$ IS DENOTED AS THE NUMBER OF SUPERPIXEL BLOCK IN THE SMALL SCALE. THE NUMBERS IN PARENTHESES REPRESENT THE STANDARD VARIANCE OF THE ACCURACIES OBTAINED ACROSS TEN REPEATED EXPERIMENTS

Dataset	Indicator	Global-based filtering Optimal parameters			Superpixel-based filtering Large scale ($S_N = B_n$)			Superpixel-based filtering Small scale ($S_N \approx 4B_n$)		
		RF [27]	GsF [44]	GrF [40]	RF [27]	GsF [44]	GrF [40]	RF [27]	GsF [44]	GrF [40]
Indian Pines	OA	89.54(1.82)	90.23(1.50)	**90.36(2.08)**	**81.43(2.13)**	80.85(2.74)	80.58(2.74)	89.30(2.00)	90.30(1.98)	**90.60(1.57)**
	AA	88.19(3.00)	**90.73(1.44)**	90.54(1.97)	**77.34(0.94)**	76.93(2.57)	77.13(1.26)	88.49(2.87)	90.74(1.53)	**91.56(0.92)**
	Kappa	88.02(2.08)	88.84(1.70)	**88.97(2.37)**	**78.83(2.36)**	78.13(3.19)	77.86(3.16)	87.76(2.28)	88.89(2.29)	**89.21(1.86)**
Salinas Valley	OA	92.73(2.47)	93.46(1.74)	**93.89(1.84)**	94.49(0.95)	95.22(2.03)	**95.68(1.09)**	95.62(2.07)	96.55(1.28)	**96.69(1.68)**
	AA	93.81(2.32)	95.20(1.36)	**96.09(0.86)**	**97.10(0.68)**	96.15(1.36)	96.36(1.08)	96.08(1.47)	97.00(0.95)	**97.19(0.82)**
	Kappa	91.89(2.76)	92.71(1.95)	**93.17(2.07)**	93.84(1.06)	94.67(2.27)	**95.18(1.22)**	95.12(2.30)	96.16(1.43)	**96.32(1.89)**
University of Pavia	OA	86.70(2.30)	87.17(1.79)	**87.69(3.27)**	88.99(2.16)	89.20(3.31)	**89.24(1.44)**	90.12(3.65)	90.42(2.57)	**91.04(3.17)**
	AA	86.39(1.95)	86.11(1.27)	**86.59(2.93)**	82.88(2.23)	88.04(2.92)	**83.70(2.19)**	83.31(3.18)	83.56(1.98)	**85.43(2.79)**
	Kappa	82.81(2.90)	83.47(2.16)	**84.12(4.06)**	85.64(2.71)	**86.02(4.10)**	85.89(1.81)	87.05(4.51)	87.44(3.27)	**88.16(3.91)**
University of Houston	OA	76.45(1.62)	**76.96(2.03)**	76.87(2.14)	77.03(1.77)	77.12(1.35)	**78.03(3.16)**	78.47(2.76)	79.35(2.20)	**80.44(3.30)**
	AA	79.68(2.04)	80.38(1.57)	**80.82(1.80)**	79.09(1.32)	79.00(1.36)	**80.70(2.67)**	79.91(1.71)	81.32(1.25)	**83.34(2.85)**
	Kappa	74.55(1.75)	**75.11(2.20)**	75.00(2.30)	75.20(1.92)	75.30(1.46)	**76.26(3.41)**	76.76(3.00)	77.71(2.39)	**78.84(3.56)**

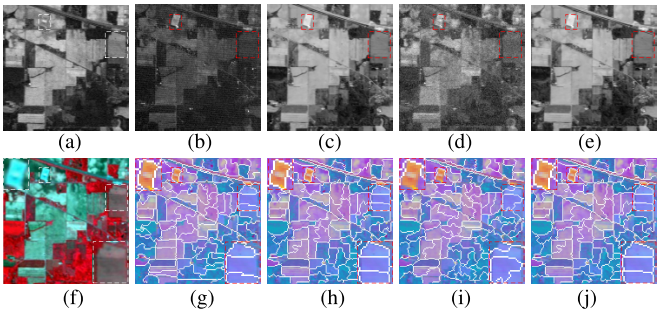


Fig. 9. Illustration of low-dimension and over-segmentation image on the Indian Pines data set. (a) One dimension (No. 111) of the Indian Pines image. The first dimension image of different dimension-reduction tools: (b) AIF, (c) PCA, (d) PPCA, and (e) MDS. (f) Three-band color composite image for Indian Pines. over-segmentation image based on different dimension-reduction tools: (g) AIF, (h) PCA, (i) PPCA, and (j) MDS.

compared to the other dimension reduction tools in terms of classification accuracy (i.e., OA, AA, and Kappa) and efficiency (run-time). Furthermore, the first dimension images of the original Indian Pines image based on different reduction dimension tools are given in Fig. 9(a)–(e), as can be observed, Fig. 9(c) and (e) have significantly less salt and pepper noise than Fig. 9(b) and (d). Meanwhile, superpixel segmentation maps based on 2-D pseudo-color images that are synthesized by the first 3-D image of the original HSI under different reduction dimension tools are studied. Fig. 9(f)–(j) shows that PCA and MDS methods-based superpixel segmentation can achieve better marginal information than the AIF and PPCA method-based segmentation results (look at the local perspective in the red rectangle). Therefore, by combining all analysis results, the MDS method is selected as a component of the proposed method and applied to subsequent experiments.

C. Analysis of Spatial Texture Consistency and Ablation

In HSI processing, filtering operations (i.e., recursive filtering (RF) [27], Gauss filtering (GsF) [44], and GrF [40]) aim to facilitate the texture consistency of spatial information that is formed by pixels with similar characteristics by enhancing the boundary information between different ground coverings and blurring the texture information within the boundaries of similar ground coverings. In this section, the above three filtering operations are introduced into the proposed 3-D-BCS framework to test texture consistency performance in terms of global and local (superpixel) aspects. Here, we adopt the classification accuracy of the 3-D-BCS to reflect their filtering performance in terms of texture consistency. In Table III, the experimental results on the different data sets about global- and superpixel-based filtering as a component of the 3-D-BCS are given. The training samples are selected randomly, which account for 1.5% (Indian Pines), 0.2% (Salinas Valley), 0.42% (University of Pavia), and 1.0% (University of Houston) of labeled data. B_n is marked as the number of superpixel block on a large scale. $4B_n$ is denoted as the number of superpixel block on a small scale. The B_n of aforesaid data sets is set to 50, 70, 225, and 255, respectively. It can be seen from Table III that, the 3-D-BCS method using superpixel in small-scale-based GrF dramatically achieves the highest classification accuracy compared to that using global-based filters on the Salinas Valley, University of Pavia, and University of Houston data sets. However, the 3-D-BCS method using filters with superpixel on large scale cannot obtain a performance advantage with respect to the 3-D-BCS with global-based filters on the Indian Pines data set. The reasons for this are 1) the filtering operation with large-scale treats the two different classes of ground covers as a single unit of execution; 2) superpixel filtering in local large scale will make the two classes of boundary too smooth. It is noteworthy that the classification performance of superpixel filtering on the proposed 3-D-BCS framework can achieve higher accuracy compared to global filtering when the number of superpixel blocks is set to $S_N = 200$ on the Indian Pines image. This means that the setting of the number of superpixel blocks plays a decisive role in the classification performance based on superpixel guidance filtering. Also, this demonstrates that superpixel-based filtering can better retain texture consistency than global-based filtering in HSI classification. Therefore,

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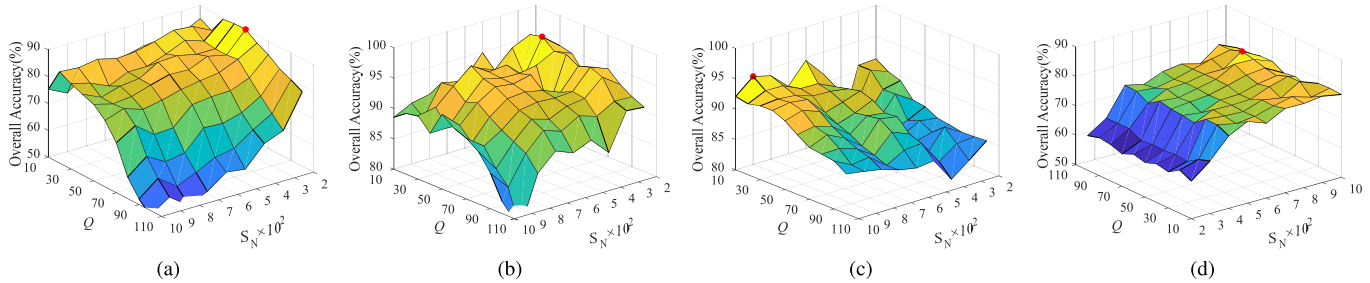


Fig. 10. Analysis of the influence of the parameters Q and S_N on the classification performance of the 3-D-BCS for different data sets. (a) Indian Pines data set (1.5% of labeled data). (b) Salinas Valley data set (0.2% of labeled data). (c) University of Pavia data set (0.42% of labeled data). (d) University of Houston data set (1% of labeled data).

superpixel-based filtering is selected in a parallel way in the proposed method for spatial information extraction.

Furthermore, another experiment is performed on the different real HSI data sets to analyze the performance contribution of each line (i.e., First Line: superpixel guidance-based 2-D Gabor, Second Line: superpixel guidance-based K-means, and Third Line: superpixel guidance-based weighted mean). The selected number of training samples and test samples is shown in Table I and each experiment is repeated ten times under random training samples. As shown in Table IV, it can be seen that the contribution of the three lines to the classification accuracy of the proposed 3-D-BCS method is not significantly different in terms of OA, AA, and kappa. However, the contribution of the first line is the smallest compared to the second and third lines to evaluate from the above indicators. The reason is that the optimal scale and direction of Gabor filters on the proposed framework have not been explored in depth in this work. In the future, the optimal scale and direction of Gabor filters in HSI applications will be the focus of our research. Generally, the improvement of classification accuracy is significant by performing a decision fusion operation on the three lines.

TABLE IV

ABLATION ANALYSIS FOR EACH STEP IN THE PROPOSED METHOD. FIRST LINE: SUPERPIXEL GUIDANCE-BASED 2-D GABOR. SECOND LINE: SUPERPIXEL GUIDANCE-BASED K-MEANS. THIRD LINE: SUPERPIXEL GUIDANCE-BASED WEIGHTED MEAN. THE NUMBERS IN PARENTHESES REPRESENT THE STANDARD VARIANCE OF THE ACCURACIES OBTAINED ACROSS TEN REPEATED EXPERIMENTS

Indian Pines				
Indicator	First Line	Second Line	Third Line	3D-BCS
OA	84.72(1.03)	**86.09(2.23)**	85.68(1.75)	87.86(1.72)
AA	**88.96(2.59)**	87.35(1.78)	87.19(1.72)	90.05(1.88)
Kappa	82.10(1.15)	**84.35(2.52)**	83.83(2.05)	86.07(1.98)
Salinas Valley				
Indicator	First Line	Second Line	Third Line	3D-BCS
OA	92.54(1.50)	93.36(0.81)	**93.51(2.38)**	94.24(1.58)
AA	92.01(0.75)	**95.08(1.28)**	93.97(1.92)	95.83(1.45)
Kappa	91.91(1.68)	93.05(0.90)	**93.32(2.68)**	93.58(1.76)
University of Pavia				
Indicator	First Line	Second Line	Third Line	3D-BCS
OA	88.32(0.74)	**89.83(2.30)**	88.74(1.36)	90.94(2.36)
AA	85.41(1.46)	**87.65(2.46)**	86.22(1.63)	88.13(3.08)
Kappa	85.60(0.97)	**87.30(2.90)**	86.82(1.72)	88.17(2.97)
University of Houston				
Indicator	First Line	Second Line	Third Line	3D-BCS
OA	75.78(1.83)	**77.24(2.38)**	76.15(2.74)	79.16(3.43)
AA	79.76(2.00)	80.45(2.46)	**81.45(1.03)**	82.50(2.65)
Kappa	73.83(1.96)	**75.33(2.56)**	74.06(2.97)	78.15(3.72)

D. Parameter Setting

In this section, the influence of parameters on the classification performance of the proposed 3-D-BCS method is analyzed, such as the number of superpixels S_N, the dimensions of the reduced image Q, the ratio of selected samples in superpixels K, and the half-peak width σ. As shown in Table I, for the Indian Pines, Salinas Valley, University of Pavia, and University of Houston data sets, the 1.5%, 0.2%, 0.42%, and 1.0% labeled data sets of the reference data are selected as training sets, respectively.

In Fig. 10, the first experiment is performed on the above real data sets to dissect the influence of paremeters S_N and Q on the proposed 3-D-BCS method. The ranges of S_N and Q are set to $2 \times 10^2 - 10 \times 10^2$ and $10 - 110$, respectively. As shown in Fig. 10(a)–(d), the OAs of the classification accuracies obtained by the proposed 3-D-BCS exhibit a close relationship with the variation of the parameters S_N and Q. Focusing on Fig. 10(a), when Q is fixed, the OAs showed a slow downward trend as S_N grew. This demonstrates that texture consistency does not depend solely on the shape and size of the superpixel block in the proposed 3-D-BCS. Analogously, OAs showed a trend of rising initially and then decreasing when Q increased while S_N was fixed. This proves that the classification accuracy of the 3-D-BCS can be promoted by overcoming the "Hughes" phenomenon of hyperspectral data. Generally, optimal parameters can be determined based on the best OAs for diverse data sets, such as the Indian Pines ($S_N = 200$; $Q = 50$), Salinas Valley ($S_N = 300$; $Q = 30$), University of Pavia ($S_N = 900$; $Q = 10$), and University of Houston ($S_N = 900$; $Q = 80$) data sets.

In the second experiment, we analyzed what affect the parameters K and σ on the classification performance of the 3-D-BCS method for four real data sets. As shown in Fig. 11, the classification results of the proposed 3-D-BCS method initially increased and then decreased as the ratio of selected samples in the superpixels increased. If the proportion of selected samples in the superpixels was large, some low-confidence samples in superpixel will be selected for the top K maximum spectral set, which will affect the quality of the K-means features. Hence, the default parameters of

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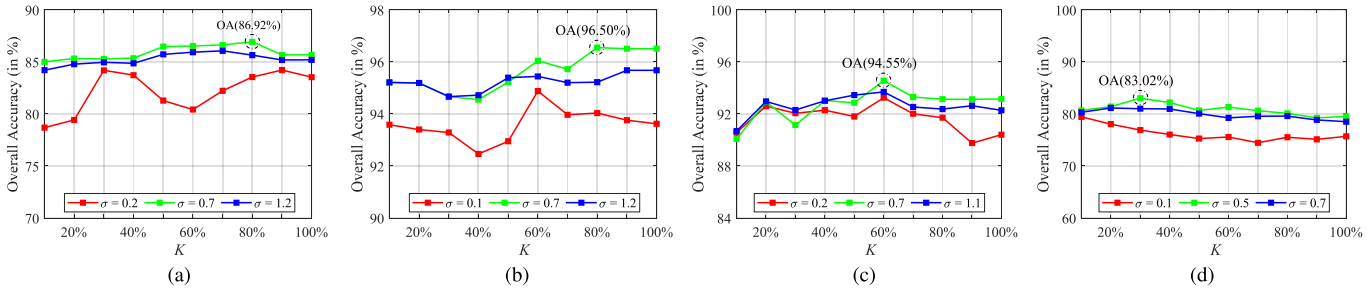


Fig. 11. Analysis of the influence of parameters K and σ on the classification performance of the 3-D-BCS for different data sets. (a) Indian Pines data set (1.5% of labeled data). (b) Salinas Valley data set (0.2% of labeled data). (c) University of Pavia data set (0.42% of labeled data). (d) University of Houston data set (1% of labeled data).

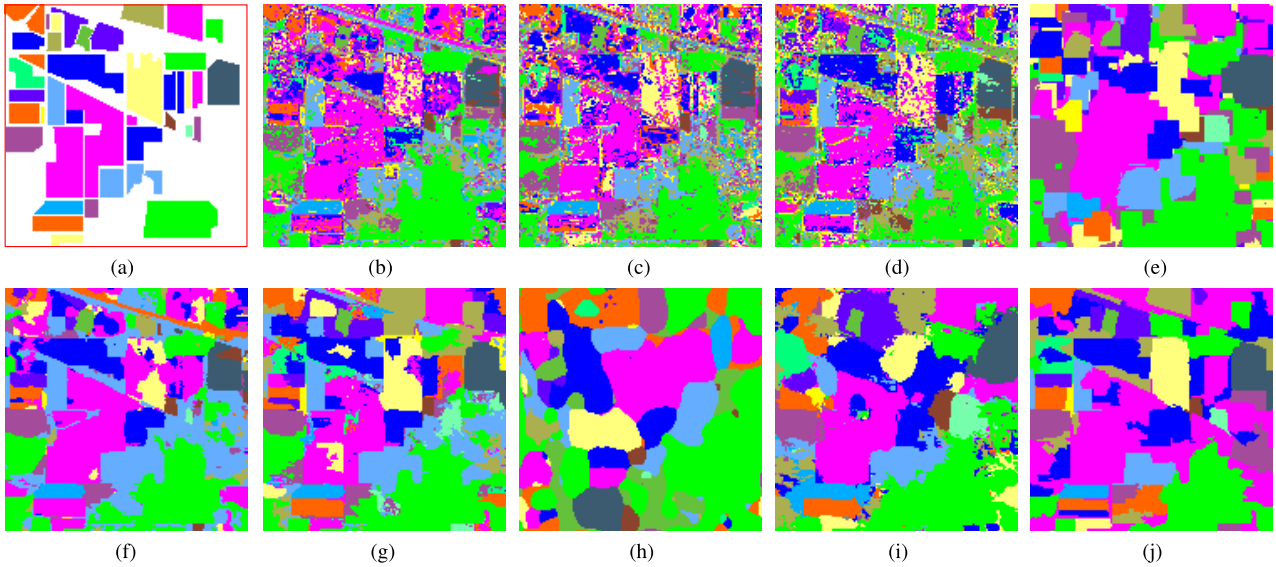


Fig. 12. Classification results (%) of various methods on the Indian Pines data set when training samples account for 1.5% of the reference data. (a) Reference data. (b) SVM (OA = 61.73%). (c) DMLSR (OA = 63.69%). (d) SaCRT (OA = 65.53%). (e) JSRC (OA = 72.90%). (f) EPF (OA = 71.49%). (g) SC-MK (OA = 81.80%). (h) WMRF (OA = 82.50%). (i) GFDN (OA = 82.82%). (j) 3-D-BCS (OA = 88.51%).

various data sets (i.e., Indian Pines, Salinas Valley, University of Pavia, and the University of Houston) are set to 80%, 80%, 60%, and 30%, respectively. Furthermore, focusing on the change of the parameter σ, similar conclusions can be reached, as the ratio K increases, the OA initially increases and then decreases. In fact, σ controls the extreme value of the Gaussian weighting function and results in a different degree of distance weight between different samples. According to the best classification accuracy, $K = 0.7$, $K = 0.7$, $K = 0.7$, and $K = 0.5$ are, respectively, set as the optimal parameters on the abovementioned real data sets. In addition, the K parameter is recommended to be set to $K = 0.7$ if the 3-D-BCS is executed on a new real data set.

E. Discussion of Various Training Ways

In this section, we conduct a group of experiments to analyze the classification performance of the proposed 3-D-BCS method using different partition ways for generating the training and test sets on the Indian Pines data set with 1.5% and 6% of labeled data. As shown in Fig. 13(a) and (b), a way to construct training sets and test sets is introduced in this work, in which the training and the test samples are always disjoint, and details are recorded in [21], [45]. These disjoint data sets are available from the *GRSS Data and Algorithm Standard Evaluation website*[3]. Fig. 13(c) and (d) shows a common way to generate training and test sets, the training samples are first randomly selected from the available ground truth, and then the remaining labeled samples are employed for testing.

Table V presents the classification results of the proposed 3-D-BCS method using different partition ways. It can be seen that the 3-D-BCS using training set generated by random way achieves the highest classification accuracies with respect to that using training set created by the disjoint way in terms of OA, AA, Kappa. And this advantage becomes more significant as the training sample increases. As shown in Fig. 14, in addition to the inferiority of the classification accuracy, the 3-D-BCS using training set created by the disjoint way exists a large number of over-smoothing and misclassification phenomena in the final classification results. This means that the 3-D-BCS method trained using the training set constructed by the disjoint way has weak discriminability for the test

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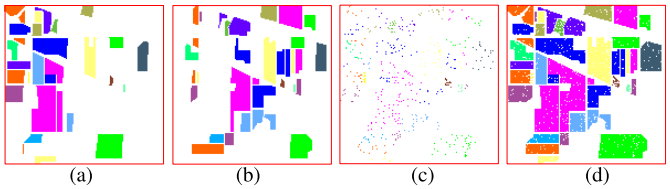


Fig. 13. Illustration of different train/test splits into the Indian Pines data set. (a) and (b) are training and testing samples in spatially disjoint way, which both account for 50% of the labeled data. (c) and (c) are training and testing samples in randomly selected way, which accounted for 6% and 94% of labeled data, respectively.

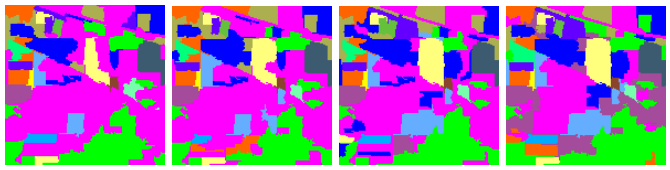


Fig. 14. Classification results (%) of the proposed 3-D-BCS method using different selection ways of training set on the Indian Pines. (a) and (b) Spatially disjoint way, and training samples accounted for 1.5% and 6% of the labeled samples, respectively. (c) and (d) Randomly selected way, and training samples accounted for 1.5% and 6% of the labeled samples, respectively.

TABLE V

CLASSIFICATION ACCURACIES (IN %) OF THE 3-D-BCS METHOD USING DIFFERENT SELECTION WAYS OF TRAINING SET ON THE INDIAN PINES DATA SET. THE NUMBERS IN PARENTHESES REPRESENT THE STANDARD VARIANCE OF THE ACCURACIES OBTAINED ACROSS TEN REPEATED EXPERIMENTS

Indicator	Spatially disjoint		Randomly selected	
	1.5%	6.0%	1.5%	6.0%
OA	64.92(4.74)	71.05(3.00)	88.40(1.77)	95.38(0.78)
AA	62.58(4.22)	69.10(2.93)	88.65(2.84)	93.27(0.61)
Kappa	61.25(6.08)	66.41(3.34)	86.71(2.04)	94.71(0.89)

samples. The reason is that the 3-D-BCS method employs very few training samples to define the discriminant rules that have low versatility when using spatially disjoint way. In the future, the improvement of classification performance on the 3-D-BCS method trained using the training set constructed by the disjoint way will be the research topic we need to focus on. Taking into account the universality of the proposed 3-D-BCS method, the random strategy for constructing the training and test sets is adopted to perform the subsequent experiments.

F. Comparison of Different Methods

In this section, the classification performance of the proposed 3-D-BCS method is compared with several widely used classification methods such as pixel-level classifiers (i.e., SVM [12], discriminative marginalized least-squares regression (DMLSR) [46], and structure-aware collaborative representation with Tikhonov regularization (SaCRT) [47] methods), spectral–spatial classification methods (i.e., joint-SRC (JSRC) [28], EPF [30], and superpixel-based classification via multiple kernels (SC-MK) [48] methods), and state-of-the-art method (i.e., weighted Markov random fields (WMRF) [49], GrF-based deep network (GFDN) [50], and 3-D-BCS methods). The SVM method is implemented in the LIBSVM library, and the parameters are set based on the five-layer cross validation and Gaussian kernel. The default parameters given in [47] are adopted for the DMLSR method.

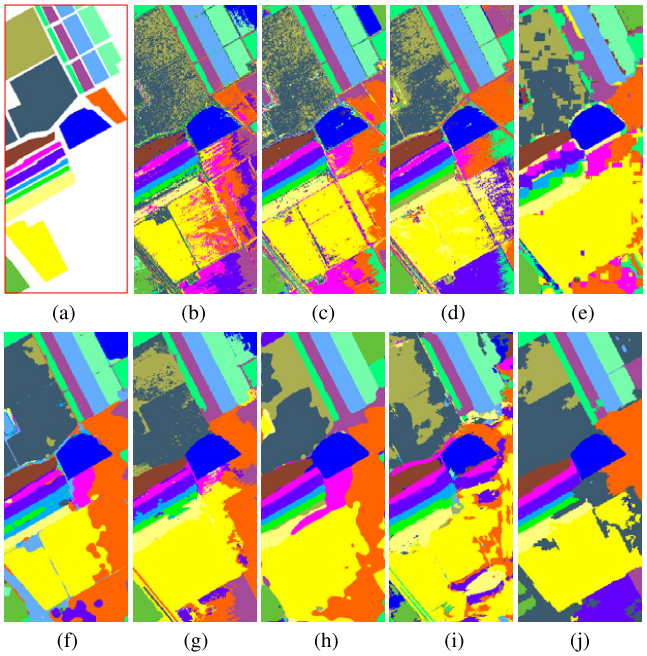


Fig. 15. Classification results (%) of different methods on the Salinas Valley data set when training samples accounted for 0.2% of the reference data. (a) Reference data. (b) SVM (OA = 81.91%). (c) DMLSR (OA = 83.51%). (d) SaCRT (OA = 85.98%). (e) JSRC (OA = 83.66%). (f) EPF (OA = 87.94%). (g) SC-MK (OA = 92.96%). (h) WMRF (OA = 90.50%). (i) GFDN (OA = 86.17%). (j) 3-D-BCS (OA = 93.55%).

The implementations provided by the authors and the default parameters are adopted for the SaCRT and JSRC. The software and the default parameters provided by the authors are employed for the EPF, SC-MK, and WMRF methods. In the proposed 3-D-BCS method, the default parameters obtained using the above parameter analysis experiment were adopted in the following experiments. To objectively evaluate the effectiveness of different classification methods, the classification experiment was repeated ten times to obtain the average value and standard deviation of OA, AA, and Kappa. The numbers in parentheses represent the standard deviation of the accuracy obtained in repeated experiments, and the best results for each class are highlighted in bold.

The first experiment was performed on the Indian Pines data set. In this, 1.5% of the reference data was used to label the data training samples (see Table I), and Fig. 12 and Table VI show the classification results and accuracies of the different competitive methods. As shown in Fig. 12, the SVM, SaCRT, and DMLSR classifiers only consider the spectral information and the classification results obtained by the above classifiers generate more misclassification phenomenon, i.e., noisy estimations. For the spectral–spatial classification methods, such as JSRC, EPF, SC-MK, WMRF, and GFDN, their classification results are smoother than the classifiers that only consider the spectral information. By contrast, the proposed 3-D-BCS method only presents a few noisy estimations in the classification map. Table VI presents the classification accuracies of diverse methods. As can be observed from Table VI, the proposed 3-D-BCS method outperforms the other compared methods for the most classes in terms of classification accuracy. For instance, the classification accuracy of the *Alfalfa*, *Hay-W*, *Oats*, and *Wheat* classes reaches 100%

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TABLE VI

CLASSIFICATION ACCURACIES (IN %) OF THE SVM [12], DMLSR [46], SACRT [47], JSRC [28], EPF [30], SC-MK [48], WMRF [49], GFDN [50], AND 3D-BCS METHODS ON THE INDIAN PINES DATA SET WITH 1.5% TRAINING SAMPLES, THE HIGHEST ACCURACIES ARE PRESENTED IN BOLD

Class	Pixel-level classifier			Spectral-spatial classification method			State-of-the-art method		
	SVM	DMLSR	SaCRT	JSRC	EPF	SC-MK	WMRF	GFDN	3D-BCS
1	29.66(8.60)	87.62(11.1)	91.90(3.19)	94.76(5.71)	67.78(27.3)	97.86(0.71)	89.41(8.73)	99.05(1.30)	**100.0(0.00)**
2	51.34(9.55)	44.73(4.99)	58.12(11.9)	68.83(2.59)	63.28(5.77)	85.70(4.82)	82.65(8.19)	75.15(11.7)	**89.95(6.78)**
3	49.87(10.9)	48.60(13.0)	44.99(4.01)	61.27(11.8)	74.02(19.9)	75.03(5.68)	61.28(3.17)	78.12(10.9)	**83.22(10.7)**
4	45.87(10.3)	41.63(7.35)	57.00(14.6)	70.90(20.3)	53.27(26.6)	73.73(13.5)	51.69(15.1)	94.25(6.74)	**95.33(9.33)**
5	72.35(9.47)	61.39(19.8)	86.09(8.92)	82.35(6.70)	94.46(5.28)	69.60(6.14)	78.04(7.22)	78.99(12.4)	**97.80(4.40)**
6	77.19(3.40)	84.29(7.08)	84.01(8.66)	77.21(8.62)	76.55(6.51)	92.66(8.15)	97.35(1.14)	75.05(14.7)	**98.75(0.17)**
7	37.11(17.1)	71.54(25.3)	74.62(26.7)	**100.0(0.00)**	85.65(23.6)	98.80(1.83)	60.83(6.77)	**100.0(0.00)**	38.57(3.34)
8	95.04(1.90)	84.27(8.94)	91.33(5.50)	75.94(5.67)	99.06(1.41)	91.35(13.1)	99.88(0.10)	94.59(8.39)	**100.0(0.00)**
9	20.52(8.41)	78.82(12.2)	89.41(12.8)	90.59(18.8)	49.95(39.6)	**100.0(0.00)**	32.22(24.1)	96.47(5.26)	**100.0(0.00)**
10	53.99(5.40)	59.38(2.71)	49.71(10.5)	71.08(8.97)	71.12(10.2)	73.54(5.88)	53.97(16.6)	74.39(5.69)	**91.08(7.29)**
11	60.09(4.26)	63.83(3.90)	70.47(6.80)	77.52(4.83)	67.67(3.79)	87.22(5.91)	**91.52(4.25)**	89.14(3.97)	78.28(5.12)
12	41.13(11.0)	32.27(4.45)	47.50(16.3)	48.14(3.52)	63.35(21.5)	60.89(11.4)	60.30(8.20)	80.24(9.59)	**96.59(4.27)**
13	82.38(6.68)	93.37(3.33)	98.12(2.58)	88.81(5.06)	95.59(3.79)	99.85(0.23)	**100.0(0.00)**	95.05(6.79)	**100.0(0.00)**
14	89.61(2.96)	87.96(6.33)	85.43(8.42)	91.59(4.32)	87.76(2.79)	91.60(6.83)	**99.61(0.46)**	97.46(1.47)	97.01(2.18)
15	35.83(13.0)	27.24(6.05)	47.72(8.60)	52.49(7.30)	75.62(21.0)	72.26(23.1)	30.43(17.6)	82.78(12.2)	**94.84(7.88)**
16	98.36(2.78)	91.91(4.16)	91.46(5.48)	95.06(4.75)	90.49(4.42)	97.42(1.43)	90.65(4.59)	94.16(6.62)	**98.88(0.35)**
OA	60.96(2.70)	62.06(0.83)	67.61(2.26)	73.98(2.25)	72.26(1.46)	83.05(2.57)	80.52(3.01)	84.30(3.16)	**88.14(1.68)**
AA	58.77(1.31)	66.18(2.15)	72.99(1.45)	77.91(1.86)	75.98(2.99)	85.47(2.01)	73.74(1.14)	87.80(3.33)	**91.27(0.73)**
Kappa	55.11(3.23)	56.81(1.13)	63.12(2.54)	70.30(2.56)	67.92(1.62)	80.68(2.86)	77.50(3.53)	82.04(3.83)	**86.33(1.98)**

TABLE VII

CLASSIFICATION ACCURACIES (IN %) OF THE SVM [12], DMLSR [46], SACRT [47], JSRC [28], EPF [30], SC-MK [48], WMRF [49], GFDN [50], AND 3-D-BCS METHODS ON THE SALINAS VALLEY DATA SET WITH 0.2% TRAINING SAMPLES; THE HIGHEST ACCURACIES ARE PRESENTED IN BOLD

Class	Pixel-level classifier			Spectral-spatial classification method			State-of-the-art method		
	SVM	DMLSR	SaCRT	JSRC	EPF	SC-MK	WMRF	GFDN	3D-BCS
1	97.08(4.85)	97.92(1.22)	98.69(1.39)	99.14(0.93)	**100.0(0.00)**	93.29(7.79)	98.81(1.09)	93.60(7.11)	**100.0(0.00)**
2	98.15(0.82)	95.25(4.38)	98.34(1.42)	99.83(0.23)	99.67(0.61)	**100.0(0.00)**	**100.0(0.00)**	90.75(7.91)	96.10(6.64)
3	81.75(5.54)	85.81(2.14)	87.00(8.93)	97.13(4.37)	93.04(3.02)	98.36(2.50)	90.10(8.34)	76.94(13.6)	**99.70(0.26)**
4	96.10(3.02)	98.95(0.30)	99.09(0.78)	80.04(4.61)	97.77(0.11)	92.14(6.57)	**97.92(3.73)**	86.84(15.1)	95.85(1.11)
5	94.87(4.12)	93.81(3.78)	91.55(11.5)	92.56(9.89)	99.55(1.11)	96.17(3.40)	94.81(6.62)	92.80(6.26)	**99.78(0.13)**
6	99.77(0.57)	99.66(0.07)	99.56(0.16)	99.65(0.28)	99.98(0.06)	99.76(0.05)	**100.0(0.00)**	95.39(3.40)	**100.0(0.00)**
7	95.21(1.82)	99.19(0.17)	99.75(0.05)	97.05(4.92)	94.84(3.07)	99.86(0.06)	99.87(0.13)	95.04(5.44)	**99.97(0.00)**
8	63.93(4.87)	69.77(3.59)	79.05(4.94)	47.94(8.07)	70.94(11.9)	83.88(7.97)	**87.04(3.36)**	85.66(6.51)	86.80(5.41)
9	97.20(2.36)	97.23(0.81)	98.93(0.25)	96.75(2.21)	97.37(3.94)	99.99(0.01)	**100.0(0.00)**	99.08(0.79)	99.09(0.68)
10	84.49(7.33)	69.77(7.20)	80.67(6.29)	93.96(1.49)	94.23(5.53)	75.08(13.9)	81.92(10.5)	86.01(11.8)	**98.73(2.33)**
11	90.10(3.64)	90.13(3.63)	93.12(3.92)	**99.89(0.16)**	89.53(18.9)	93.90(9.10)	99.63(0.26)	84.42(14.6)	97.48(7.55)
12	84.27(8.89)	98.67(1.44)	90.75(16.5)	91.08(1.80)	92.15(9.58)	89.96(8.34)	**98.85(2.98)**	80.31(10.8)	92.72(4.79)
13	85.93(8.75)	98.36(0.54)	98.12(1.67)	88.40(4.34)	94.99(5.73)	94.02(10.6)	**99.73(0.25)**	73.52(15.2)	84.18(11.4)
14	83.99(13.2)	88.19(2.90)	94.17(3.00)	**99.24(0.84)**	98.37(2.04)	84.89(13.7)	98.41(2.86)	81.60(11.4)	94.16(11.6)
15	64.59(13.4)	54.04(3.08)	51.91(8.37)	82.27(7.00)	60.43(4.05)	84.41(4.47)	67.64(16.4)	76.90(7.87)	**96.40(4.41)**
16	98.04(2.78)	70.80(12.0)	94.14(3.16)	92.22(6.30)	**100.0(0.00)**	87.53(16.6)	87.16(12.5)	69.88(16.7)	**100.0(0.00)**
OA	82.10(1.66)	82.57(0.44)	85.95(1.01)	83.97(1.97)	87.88(5.37)	91.02(2.07)	90.62(1.89)	87.06(1.81)	**94.94(1.54)**
AA	88.47(1.28)	87.97(0.57)	90.93(1.09)	91.07(1.52)	92.68(4.48)	92.08(1.85)	93.87(1.59)	85.55(3.01)	**96.31(1.24)**
Kappa	79.96(1.95)	80.59(0.47)	84.34(1.13)	82.29(2.16)	86.38(6.10)	90.01(2.29)	89.53(2.13)	85.48(2.07)	**94.34(1.74)**

when the training sample is only 1.5% of the reference data. However, Table VI shows that the classification accuracy of *Grass-P-M* is lower than other methods. The reason is that the *Grass-P-M* class has very little labeled data (only 25), which may cause these pixels be classified as other nonbreed superpixels on segmentation operations, and lead to obtaining lower classification results. Specifically, comparing the competitive methods in Table VI shows that the proposed 3-D-BCS method achieves the highest classification performance in terms of OA, AA, and Kappa; this demonstrates that the effectiveness of the 3-D-BCS feature extraction method in the HSI classification application.

The second experiment was carried out on the Salinas Valley data set. 0.2% of the reference data was selected as a training sample and the remaining labeled data was used as the testing set. Table VII presents the classification accuracies of different classification methods, it shows that the proposed 3-D-BCS method obtains the highest accuracies for eight different classes. For example, the classification accuracies of *weeds_1*, *Stubble*, and *Vinyard_T* are 100% compared to other classification methods. Moreover, Fig. 15 shows the classification results obtained by different methods. As shown in this figure, by taking the spatial information into account, the EPF and WMRF methods can effectively improve the classification accuracy with respect to pixel-level classifiers. However, some noisy estimations of ground coverings remain visible in the obtained classification results. By contrast, the classification result using the proposed method is closer to the

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TABLE VIII

CLASSIFICATION ACCURACIES (IN %) OF THE SVM [12], DMLSR [46], SACRT [47], JSRC [28], EPF [30], SC-MK [48], WMRF [49], GFDN [50], AND 3-D-BCS METHODS ON THE UNIVERSITY OF PAVIA DATA SET WITH 0.42% TRAINING SAMPLES, THE HIGHEST ACCURACIES ARE PRESENTED IN BOLD

Class	Pixel-level classifier			Spectral-spatial classification method			State-of-the-art method		
	SVM	DMLSR	SaCRT	JSRC	EPF	SC-MK	WMRF	GFDN	3D-BCS
1	93.43(2.68)	63.23(3.14)	66.07(9.66)	33.86(6.39)	83.24(5.26)	**93.71(4.65)**	84.94(6.40)	83.72(6.57)	89.43(5.93)
2	90.38(1.50)	77.54(4.62)	75.03(4.70)	79.62(4.45)	94.41(2.49)	85.04(3.75)	81.58(7.26)	87.71(5.56)	**97.55(1.35)**
3	56.39(6.79)	71.59(2.43)	77.41(4.97)	86.27(4.55)	91.50(9.43)	84.13(2.16)	83.87(10.4)	**91.90(6.85)**	82.01(6.37)
4	65.76(13.1)	90.91(1.20)	94.11(2.80)	83.23(2.51)	66.42(16.7)	**96.81(2.12)**	90.87(4.62)	89.29(4.06)	75.23(8.39)
5	93.78(4.00)	99.26(0.46)	99.68(0.11)	96.73(3.50)	99.34(1.04)	99.70(0.11)	**100.0(0.00)**	99.64(0.54)	99.46(1.03)
6	50.16(7.90)	59.29(6.24)	67.28(5.77)	85.92(4.50)	60.33(12.7)	79.56(3.66)	84.64(9.10)	89.26(7.26)	**92.50(4.08)**
7	54.34(7.48)	86.00(1.90)	76.35(5.39)	95.49(1.86)	**100.0(0.00)**	83.52(4.94)	98.34(1.04)	98.99(2.02)	94.91(8.84)
8	75.64(5.73)	67.23(8.82)	67.25(8.91)	79.65(7.72)	**94.22(4.05)**	91.07(4.79)	86.74(8.05)	92.66(2.43)	89.72(3.39)
9	99.88(0.09)	97.28(1.60)	92.05(4.15)	26.55(6.48)	99.00(0.97)	**100.0(0.00)**	57.54(16.9)	97.18(4.20)	85.82(8.03)
OA	75.25(2.63)	74.32(1.90)	74.72(2.63)	73.71(1.60)	82.72(4.24)	87.79(2.23)	84.25(3.40)	88.94(2.65)	**91.83(1.76)**
AA	75.53(1.62)	79.15(1.14)	79.47(2.16)	74.15(1.14)	87.61(2.49)	90.39(1.41)	85.39(2.29)	**92.26(1.49)**	89.62(1.99)
Kappa	68.74(2.93)	66.97(2.13)	67.81(3.08)	66.61(1.83)	78.02(5.11)	84.12(2.85)	79.70(4.06)	85.86(3.10)	**89.29(2.22)**

TABLE IX

CLASSIFICATION ACCURACIES (IN %) OF THE SVM [12], DMLSR [46], SACRT [47], JSRC [28], EPF [30], SC-MK [48], WMRF [49], GFDN [50], AND 3-D-BCS METHODS ON THE UNIVERSITY OF HOUSTON DATA SET WITH 1% TRAINING SAMPLES; THE HIGHEST ACCURACIES ARE PRESENTED IN BOLD

Class	Pixel-level classifier			Spectral-spatial classification method			State-of-the-art method		
	SVM	DMLSR	SaCRT	JSRC	EPF	SC-MK	WMRF	GFDN	3D-BCS
1	89.02(5.35)	88.04(8.98)	81.60(6.72)	**92.86(5.55)**	91.19(6.15)	79.68(10.6)	90.10(5.30)	72.77(6.10)	77.48(2.61)
2	90.40(5.27)	**93.50(4.60)**	89.98(4.36)	85.14(4.61)	89.80(6.79)	64.65(7.78)	90.00(4.64)	65.18(11.5)	60.06(5.89)
3	89.28(7.37)	99.71(0.10)	99.59(0.28)	85.90(3.00)	93.68(5.95)	**100.0(0.00)**	99.84(0.10)	97.67(3.51)	**100.0(0.00)**
4	**94.78(4.57)**	86.71(4.84)	87.65(5.60)	87.43(3.82)	91.80(9.54)	77.52(10.5)	92.79(2.91)	64.25(13.2)	58.80(8.50)
5	89.76(3.15)	98.46(0.41)	98.54(1.00)	98.44(1.26)	92.80(2.84)	82.09(10.0)	97.31(3.11)	85.30(8.35)	**99.21(4.07)**
6	82.50(19.6)	87.87(5.92)	91.81(4.58)	80.63(6.60)	85.30(20.7)	94.16(4.54)	88.44(5.06)	**95.02(5.32)**	79.77(13.3)
7	72.12(9.04)	61.16(6.29)	74.64(8.94)	42.00(5.84)	**79.57(6.58)**	78.05(5.40)	70.22(7.23)	73.02(9.42)	73.43(7.92)
8	71.56(9.93)	56.27(14.5)	59.40(6.74)	58.25(6.55)	75.20(10.1)	47.75(8.58)	58.70(8.44)	44.81(8.89)	**74.62(4.04)**
9	64.22(6.93)	59.52(11.3)	50.82(13.7)	65.88(6.64)	75.33(11.5)	65.59(8.44)	63.41(7.84)	53.63(12.3)	**78.76(4.87)**
10	61.98(7.77)	82.10(3.97)	83.81(2.02)	85.47(8.89)	69.73(9.06)	74.26(9.43)	74.82(6.36)	85.94(9.79)	**87.10(4.01)**
11	60.63(3.81)	57.03(4.77)	56.56(10.3)	73.32(8.76)	68.67(6.91)	81.18(8.46)	54.55(8.40)	74.19(9.55)	**86.95(4.85)**
12	60.08(10.1)	61.11(4.54)	54.33(11.7)	61.73(8.49)	64.50(13.9)	56.18(5.82)	49.30(9.71)	51.57(6.39)	**83.73(5.48)**
13	30.66(11.2)	34.47(3.23)	44.84(5.12)	46.54(13.1)	30.35(14.3)	67.54(6.14)	46.62(8.50)	68.15(9.25)	**80.87(14.1)**
14	80.54(9.24)	89.28(17.1)	98.95(0.80)	99.69(0.39)	84.21(11.7)	**100.0(0.00)**	98.47(0.71)	96.03(5.48)	**100.0(0.00)**
15	99.00(0.70)	98.52(0.85)	98.71(0.23)	99.38(0.68)	99.82(0.42)	99.86(0.05)	98.58(0.40)	97.38(4.03)	**100.0(0.00)**
OA	74.91(3.00)	76.08(1.97)	76.23(1.91)	76.50(1.80)	78.73(3.12)	74.55(2.93)	76.55(1.05)	71.23(3.19)	**79.82(1.31)**
AA	75.77(2.65)	76.92(1.80)	78.08(1.60)	77.51(1.75)	79.46(2.46)	77.90(2.36)	78.21(0.95)	74.99(2.77)	**82.72(1.91)**
Kappa	72.90(3.22)	74.15(2.12)	74.31(2.06)	74.59(1.95)	77.02(3.35)	72.53(3.16)	74.65(1.14)	68.88(3.47)	**78.19(1.40)**

reference data with respect to the above-mentioned methods. This proves the superiority of the proposed 3-D-BCS method in HSI classification.

Another experiment was conducted on the University of Pavia and the University of Houston real data sets. For the University of Pavia and the University of Houston data sets, 0.42% and 1% of the labeled data, respectively, were randomly selected as the training set. As shown in Tables VIII and IX, it can be seen that the proposed 3-D-BCS still achieved the highest classification performance with respect to other competitive methods in terms of OA, AA, and Kappa. This further demonstrates that the generalization ability of the proposed 3-D-BCS method in terms of feature extraction for HSI classification tasks of various real scenes. Furthermore, to test the robustness of the 3-D-BCS method, another experiment is conducted on the Indian Pines, Salinas Valley, University of Pavia, and University of Houston data sets with various training sets. The rates referring to the randomly selected training samples that account for a given percentage of the ground truth, are selected from the interval of $\{0.50\%, 1.00\%, \ldots, 5.00\%\}$, $\{0.40\%, 0.60\%, \ldots, 2.00\%\}$, $\{0.21\%, 0.42\%, \ldots, 2.10\%\}$, and $\{0.50\%, 1.00\%, \ldots, 5.00\%\}$, respectively. As shown in Fig. 16, the 3-D-BCS always achieves the highest classification results in terms of OAs with respect to competitive methods on all of the real HSI data set. Specifically, compared with the deep learning method (i.e., GFDN), the proposed method has the same classification advantage in the case of limited training samples. Therefore, this experiment proves the practicability of the 3-D-BCS in the HSI classification application.

G. Computational Complexity

For the above-mentioned experiments, all experimental programs were implemented on a laptop computer with a 2.30 GHz Intel Core i5-6300HQ CPU and 8 GB of RAM, and the software platform was MATLAB R2017a (MathWorks, Natick, Massachusetts, America). The run times

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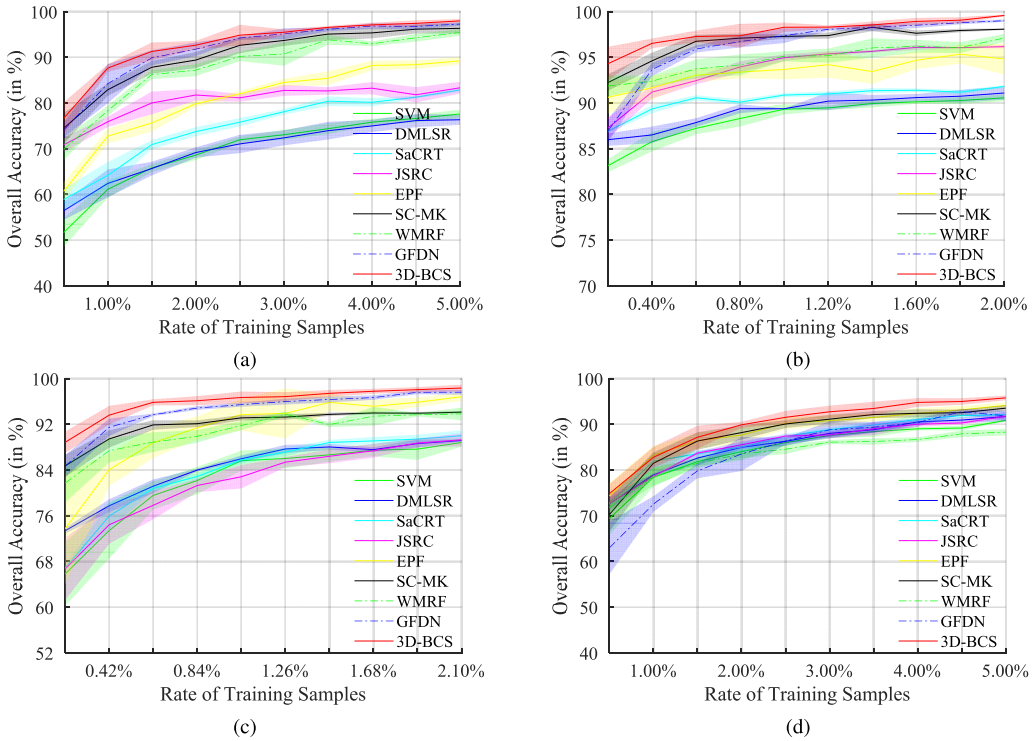


Fig. 16. Performance comparison of various competitive methods and the proposed 3-D-BCS method for the various data sets with different rates of labeled data. (a) Indian Pines. (b) Salinas Valley. (c) University of Pavia. (d) University of Houston.

TABLE X
COMPUTATION TIME (IN SECONDS) FOR COMPARISON METHODS INCLUDING SVM, DMLSR, SACRT, JSRC, EPF, SC-MK, WMRF, GFDN, AND THE PROPOSED 3-D-BCS METHOD

Methods	Pixel-level classifiers			
	Indian Pines	Salinas Valley	University of Pavia	University of Houston
SVM	5.43	4.40	13.91	17.79
DMLSR	2.57	2.29	1.97	2.20
SaCRT	36.10	181.47	63.87	41.48
Methods	Spectral-spatial classification methods			
	Indian Pines	Salinas Valley	University of Pavia	University of Houston
JSRC	53.19	296.06	135.41	23.23
EPF	3.08	33.60	8.75	22.25
SC-MK	1.49	3.45	3.34	17.87
Methods	State-of-the-art methods			
	Indian Pines	Salinas Valley	University of Pavia	University of Houston
WMRF	15.40	52.94	46.74	8.10
GFDN	171.90	177.27	110.49	179.35
3D-BCS	5.38	14.79	24.66	133.38

of the 3-D-BCS method and other competitive methods are presented for different real HSI data sets with the above training and testing set. As shown in Table X, the proposed methods could not achieve the fastest execution time with respect to most of the comparison methods. The reason for this is that the 2-D Gabor filters and superpixel segmentation take up most of the running time. However, for the same real HSI data set, the above operation only needs to be performed once in the proposed method. Thus, designing solutions to implement the proposed method more efficiently could be the focus of future research. Furthermore, the proposed method with C++ programming can be carried out in real-time. In other words, since the proposed 3-D-BCS method is computationally efficient, it will be quite suitable for solving practical problems for real applications.

V. CONCLUSION

This article proposes a novel feature extraction method via 3-D-BCS for HSI classification. Specifically, the spatial–spectral information of a local region based on a superpixel perspective is redefined to overcome the spectral–spatial weak assumptions in feature extraction. Experiments performed on several real hyperspectral data sets with limited training samples show that the proposed 3-D-BCS method outperforms other types of the classification method. However, one limitation of the proposed 3-D-BCS method is that single-scale oversegmentation and single feature representation on the 3-D-BCS is currently unable to mine the structural information of materials to the greatest extent. In the future, collaborative learning of multiscale strategy and multiple features for feature extraction will be studied. Furthermore, developing a classification method with exploiting deep structural features to further improve the quality of the 3-D characteristics sharing block will be an interesting focus in our future research.

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data sets. In addition, the authors appreciate Dr. Wei Li and Dr. Xudong Kang for providing the software of DMLSR, SaCRT, GFDN. The authors would like also to thank the editors and anonymous reviewers for their insightful comments and suggestions, which have significantly improved this article.

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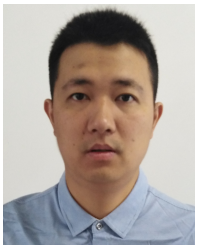
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